Integration (without limits) we add +C

$\int dx = x + C$ e.g $\int k\,dx = kx + C$

$\int 5\,dx = 5x + C$

Power Rule

$$\int x^n\,dx = \frac{x^{n+1}}{n+1} + C$$

e.g

$$\int x^2\,dx = \frac{x^3}{3} + C$$

$$\int \frac{1}{x^2}\,dx = \int x^{-2}\,dx = \frac{x^{-1}}{-1} + C$$

Derivative of trig	Antiderivatives
$\sin \to \cos$	$\sin \to -\cos$
$\cos \to -\sin$	$\cos \to \sin$
$\tan \to \sec^2 x$	$\sec^2 \to \tan$

$\sec x \tan x \rightarrow \sec x$

$\operatorname{cosec}^2 x = -\cot x$

$\operatorname{cosec} x \cot x = -\operatorname{cosec} x$

eg $\int (8\cos x + 3\sin x)\,dx$

$\Rightarrow 8\sin x - 3\cos x + C$

eg $\int [4\sec^2 x - \sec x \tan x)\,dx$

$= (4\tan x - \sec x) + C$

eg

$\int \operatorname{cosec} x \,(\cot x - \operatorname{cosec} x)\,dx$

$\int \operatorname{cosec} x \cot x - \operatorname{cosec}^2 x\,dx$

$-\operatorname{cosec} x + \cot x + C$

Integrating exponential functions.

$$\int e^u \, du = \frac{e^u}{u'} + C$$

$u' \longrightarrow$ where u is a linear function/constant

e.g $$\int e^{4x} \, dx = \frac{e^{4x}}{4} + C$$

e.g $$\int e^{7x-3} \, dx = \frac{e^{7x-3}}{7} + C$$

Logarithmic functions:

$$\int \frac{1}{u} \, du = \frac{\ln u}{u'} + C$$

where u is a linear function/constant

e.g $$\int \frac{1}{x} \, dx = \frac{\ln x}{1} + C$$

e.g. $\int \frac{1}{x+5}\,dx = \frac{\ln(x+5)}{1} + C$

e.g. $\int \frac{1}{4x-3}\,dx = \frac{\ln(4x-3)}{4} + C$